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B.Tech. Degree IV Semester Examination April 2017

IT 15-1403 SYSTEM PROGRAMMING (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain functions of an assembler.
- (b) What are assembler directives? Explain with examples.
- (c) How does the assembler handle Literals? What is the difference between Immediate operands and literals?
- (d) Explain the algorithm for an absolute loader.
- (e) Differentiate Linking loader and Linkage editor with a neat diagram.
- (f) What is macro? Differentiate a macro and a subroutine.
- (g) Explain the generation of unique labels in macro with example.
- (h) What is meant by recursive macro? How is it implemented in one pass macro processor?
- (i) Write on user interface and run-time environment.
- (j) Explain basic functions of an operating system.

PART B

(4 × 10 = 40)

- II. Write the algorithm for a 2 pass assembler with the necessary data structures needed.

OR

- III. Explain (i) Different records in an object program format.
(ii) Write a short note on machine independent assembler features.

- IV. Explain any two methods for specifying relocation as a part of object program with suitable algorithms.

OR

- V. Explain the algorithm for a linking loader with data structures.

- VI. What are the different data structures used in the implementation of a single pass macro processor? Write the algorithm for a single pass macro processor.

OR

- VII. Explain with suitable examples:
 - (i) Concatenation of macro parameters.
 - (ii) Keyword macro parameters.
 - (iii) Conditional macro expansion.

- VIII. Explain the different types of operating systems with a neat diagram.

OR

- IX. Explain (i) Virtual machine.
(ii) Hierarchical structure of an operating system.